Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

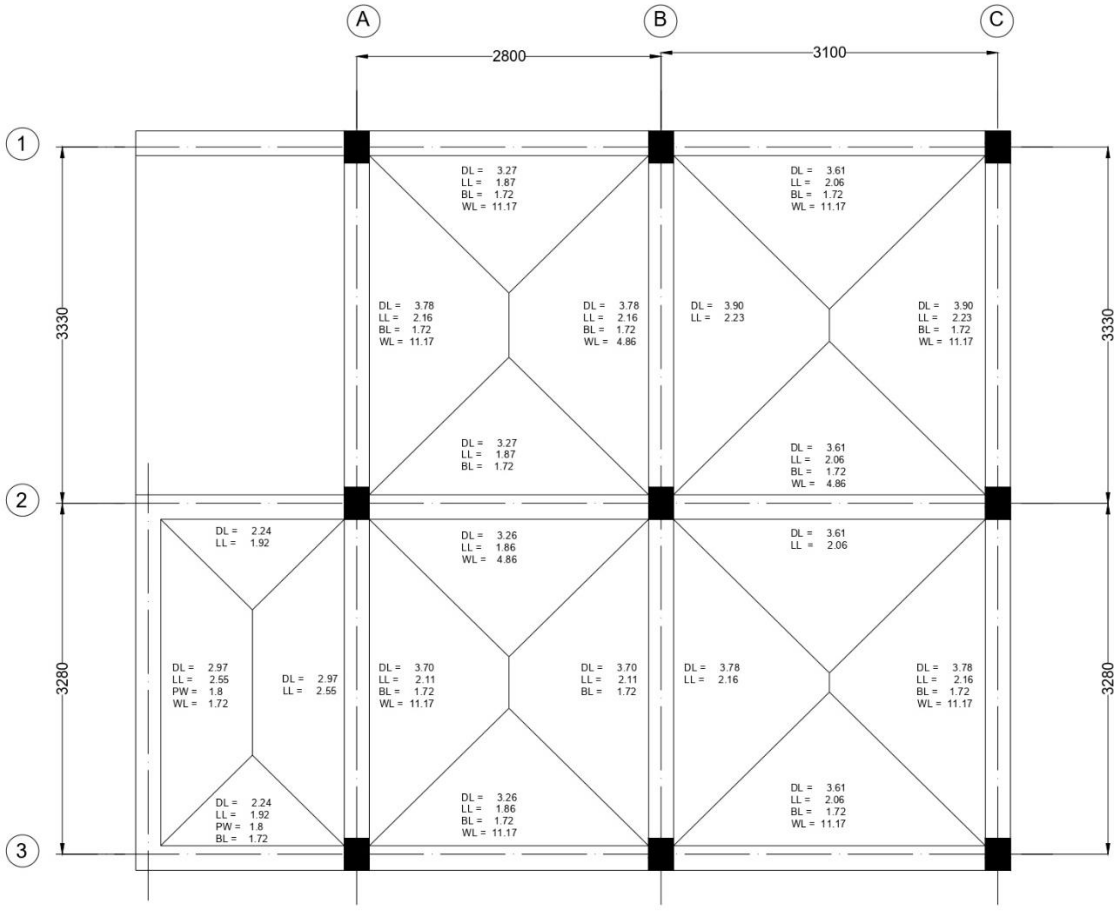


Figure 1: Beam Layout

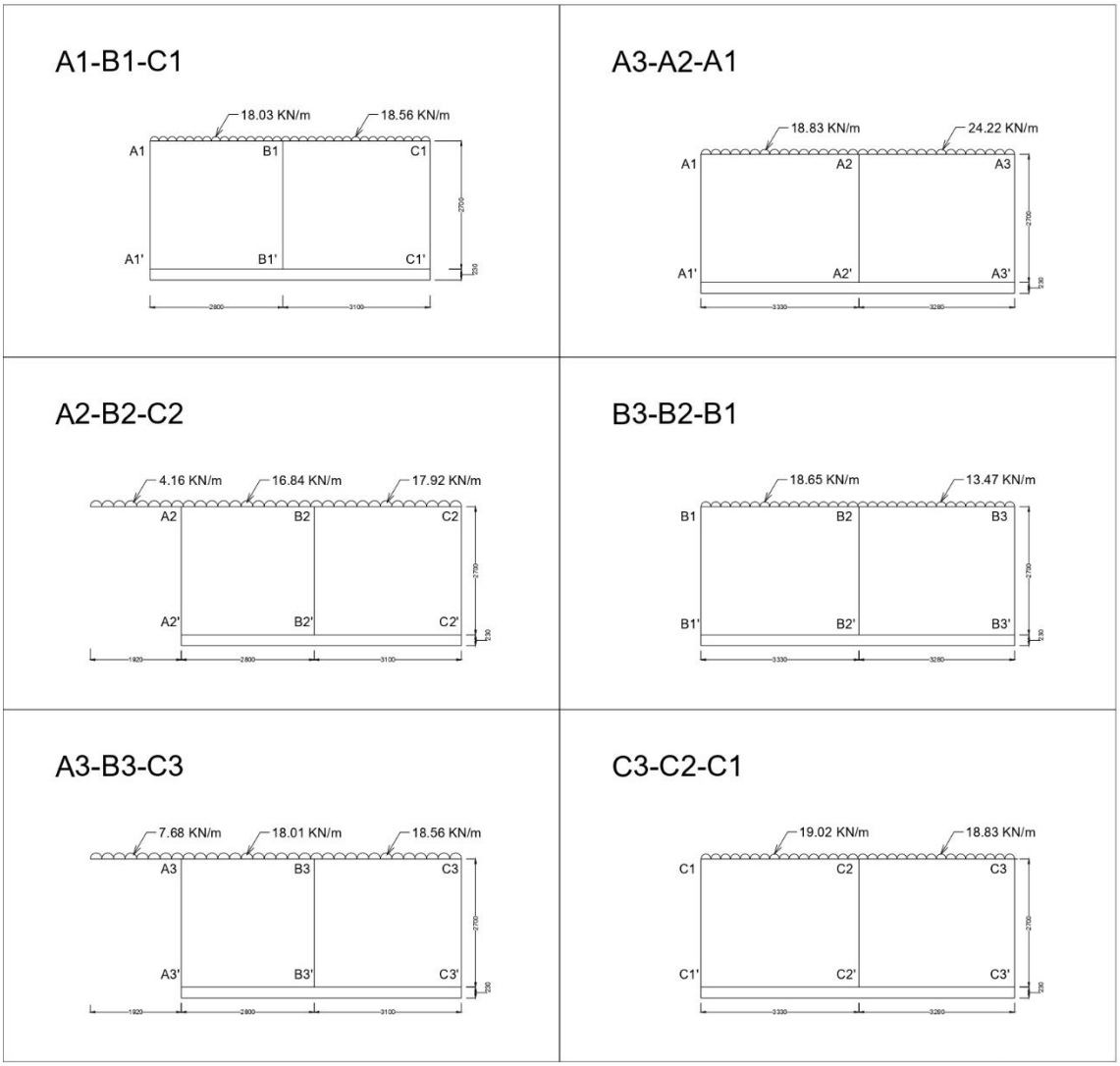
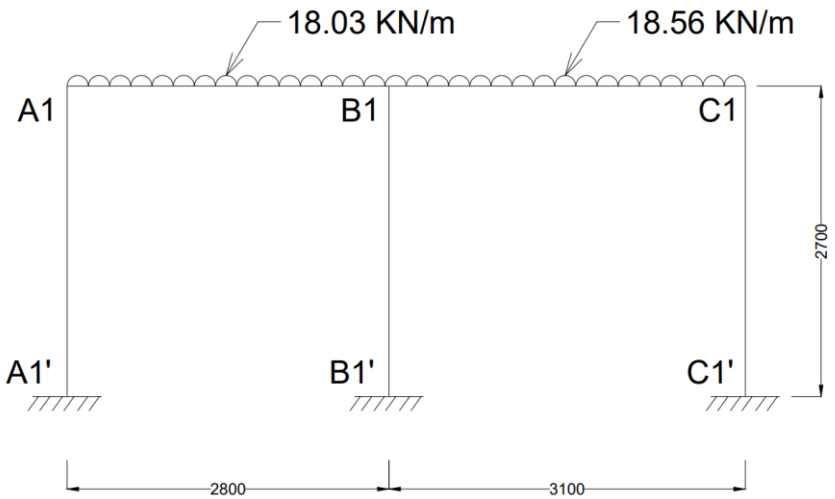


Figure 2: Frames.

Analysis of Frame A1-B1-C1



Step 1: Fixed End Moments

$MF_{A1B1} = \frac{-WL^2}{12} = \frac{-18.03 \times 2.8^2}{12}$ = -11.78 KN-m

$MF_{B1A1} = \frac{WL^2}{12} = \frac{18.03 \times 2.8^2}{12}$ = +11.78 KN-m

$MF_{B1C1} = \frac{-WL^2}{12} = \frac{-18.56 \times 3.1^2}{12}$ = -14.86 KN-m

$MF_{C1B1} = \frac{WL^2}{12} = \frac{18.56 \times 3.1^2}{12}$ = +14.86 KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{A1A1'} = MF_{A1'A1} = MF_{B1B1'} = MF_{B1'B1} = MF_{C1C1'} = MF_{C1'C1} = 0$

Step 2: Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right), \left(k = \frac{4EI}{L}\right)$

Joints	Members	Stiffness (k)	Σk	R.F
A1	A1B1	1.428EI	2.909EI	-0.245
	A1A1′	1.481EI		-0.255
B1	B1A1	1.428EI	4.200EI	-0.170
	B1C1	1.290EI		-0.154
	B1B1′	1.481EI		-0.176
C1	C1B1	1.290EI	2.771EI	-0.233
	C1C1′	1.481EI		-0.267

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A1B1} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{A1A1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B1A1} = \left(\frac{4EI}{2.8}\right) = 1.428EI$

$K_{B1C1} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{B1B1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C1B1} = \left(\frac{4EI}{3.1}\right) = 1.290EI$

$K_{C1C1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{A1B1} = -\frac{1}{2}\left(\frac{1.428EI}{2.909EI}\right) = -0.245$

$R.F_{A1A1'} = -\frac{1}{2}\left(\frac{1.481EI}{2.909EI}\right) = -0.255$

$R.F_{B1A1} = -\frac{1}{2}\left(\frac{1.428EI}{4.199EI}\right) = -0.170$

$R.F_{B1C1} = -\frac{1}{2}\left(\frac{1.290EI}{4.199EI}\right) = -0.154$

$R.F_{B1B1'} = -\frac{1}{2}\left(\frac{1.481EI}{4.199EI}\right) = -0.176$

$R.F_{C1B1} = -\frac{1}{2}\left(\frac{1.290EI}{2.771EI}\right) = -0.233$

$R.F_{C1C1'} = -\frac{1}{2}\left(\frac{1.481EI}{2.771EI}\right) = -0.267$

Sum of Stiffness $\sum k$ at Joints

A1: 1.428EI + 1.481EI = 2.909EI

B1: 1.428EI + 1.290EI + 1.481EI = 4.200EI

C1: 1.290EI + 1.481EI = 2.771EI

Step 3: Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

Storey	Members	k	Σk	D.F
1	A1A1′	1.481EI	4.444EI	-0.5
	B1B1′	1.481EI		-0.5
	C1C1′	1.481EI		-0.5

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{A1A1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B1B1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{C1C1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{A1A1'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

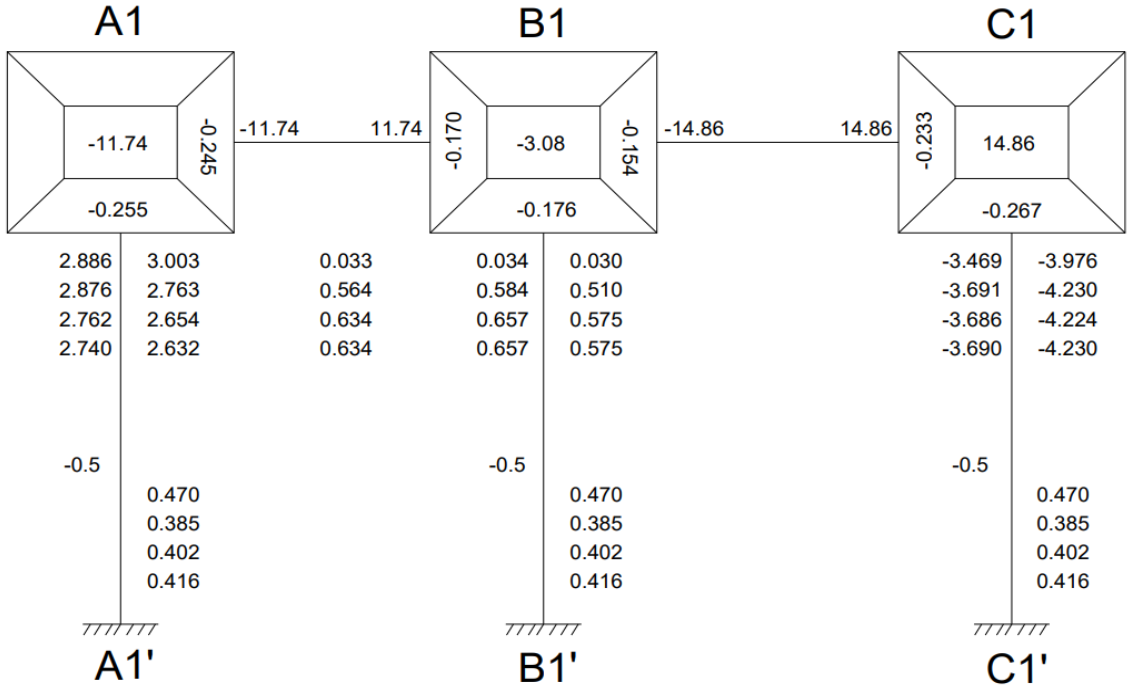
$D.F_{B1B1'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{C1C1'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

Sum of Stiffness $\sum k$ at Storey

$\sum k = 1.481EI + 1.481EI + 1.481EI = 4.444EI$

Rotation and Displacement Contribution



Cycle 1

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1B1} = -0.245 × [-11.78 + 0 + 0] = 2.886

$M_{A1A1'}$ = -0.255 × [-11.78 + 0 + 0] = 3.003

M_{B1A1} = -0.170 × [-3.08 + 2.886 + 0] = 0.033

M_{B1C1} = -0.154 × [-3.08 + 2.886 + 0] = 0.030

$M_{B1B1'}$ = -0.176 × [-3.08 + 2.886 + 0] = 0.034

M_{C1B1} = -0.233 × [14.86 + 0.030 + 0] = -3.469

$M_{C1C1'}$ = -0.267 × [14.86 + 0.030 + 0] = -3.976

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (3.003 + 0.034 + (-3.976)) = 0.470

Cycle 2

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1B1} = -0.245 × [-11.78 + 0.470 + 0.033] = 2.763

$M_{A1A1'}$ = -0.255 × [-11.78 + 0.470 + 0.033] = 2.876

M_{B1A1} = -0.170 × [-3.08 + 0.470 + 2.763 + (-3.469)] = 0.564

M_{B1C1} = -0.154 × [-3.08 + 0.470 + 2.763 + (-3.469)] = 0.510

$M_{B1B1'}$ = -0.176 × [-3.08 + 0.470 + 2.763 + (-3.469)] = 0.584

M_{C1B1} = -0.233 × [14.86 + 0.470 + 0.510] = -3.691

$M_{C1C1'}$ = -0.267 × [14.86 + 0.470 + 0.510] = -4.230

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (2.876 + 0.584 + (-4.230)) = 0.385

Cycle 3

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1B1} = -0.245 × [-11.78 + 0.385 + 0.564] = 2.654

$M_{A1A1'}$ = -0.255 × [-11.78 + 0.385 + 0.564] = 2.762

M_{B1A1} = -0.170 × [-3.08 + 0.385 + 2.654 + (-3.691)] = 0.634

M_{B1C1} = -0.154 × [-3.08 + 0.385 + 2.654 + (-3.691)] = 0.575

$M_{B1B1'}$ = -0.176 × [-3.08 + 0.385 + 2.654 + (-3.691)] = 0.657

M_{C1B1} = -0.233 × [14.86 + 0.385 + 0.575] = -3.686

$M_{C1C1'}$ = -0.267 × [14.86 + 0.385 + 0.575] = -4.224

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (2.762 + 0.657 + (-4.224)) = 0.4025

Cycle 4

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{A1B1} = -0.245 × [-11.78 + 0.4025 + 0.634] = 2.632

$M_{A1A1'}$ = -0.255 × [-11.78 + 0.4025 + 0.634] = 2.740

M_{B1A1} = -0.170 × [-3.08 + 0.4025 + 2.632 + (-3.686)] = 0.634

M_{B1C1} = -0.154 × [-3.08 + 0.4025 + 2.632 + (-3.686)] = 0.575

$M_{B1B1'}$ = -0.176 × [-3.08 + 0.4025 + 2.632 + (-3.686)] = 0.657

M_{C1B1} = -0.233 × [14.86 + 0.4025 + 0.575] = -3.690

$M_{C1C1'}$ = -0.267 × [14.86 + 0.4025 + 0.575] = -4.230

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{A1A1'} + M_{B1B1'} + M_{C1C1'}$ = -0.5 × (2.740 + 0.657 + (-4.230)) = 0.4165

Step 4: Final End Moments

M_{ij} = FEM + 2×M_{ij} (Near end Rotation) + M_{ji} (Far end Rotation) +M″ (Displacement)

Beams (No Sway M″= 0)

M_{A1B1} = -11.78 + 2(2.632) + 0.634 = -5.88

M_{B1A1} = 11.78 + 2(0.634) + 2.632 = 15.68

M_{B1C1} = -14.86+ 2(0.575) + (-3.690) = -17.40

M_{C1B1} = 14.86 + 2(-3.690) + 0.575 = 8.05

Column (With Sway M″= 0.4165)

$M_{A1A1'}$ = 0 + 2(2.740) + 0 + 0.4165 = 5.89

$M_{A1'A1}$ = 0 + 2(0) + 2.740 + 0.4165 = 3.15

$M_{B1B1'}$ = 0 + 2(0.657) + 0 + 0.4165 = 1.73

$M_{B1'B1}$ = 0 + 2(0) + 0.657 + 0.4165 = 1.07

$M_{C1C1'}$ = 0 + 2(-4.230) + 0 + 0.4165 = -8.04

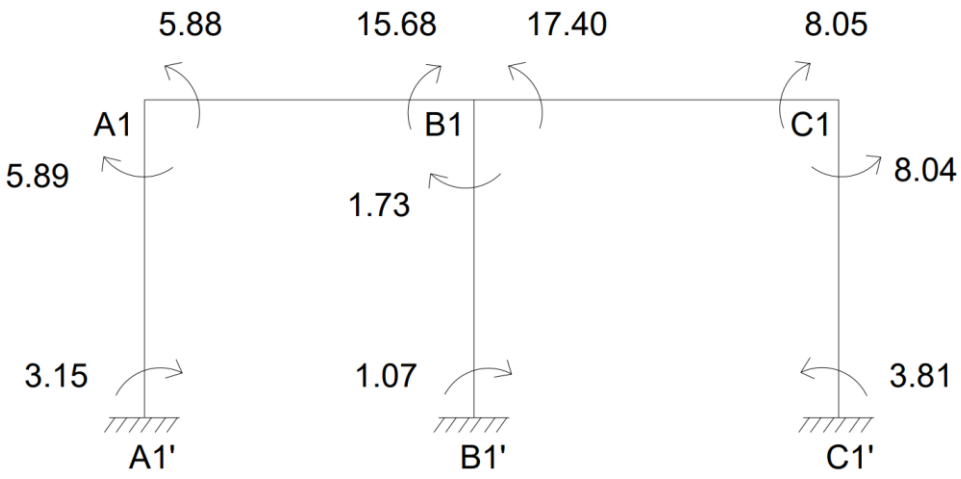
$M_{C1'C1}$ = 0 + 2(0) + (-4.230) + 0.4165 = -3.81

Equilibrium Check

Joint A1 = M_{A1B1} + $M_{A1A1'}$ = -5.88 + 5.89 = 0.01

Joint A1 = M_{B1A1} + $M_{B1C1'}$ + $M_{B1B1'}$ = 15.68 + (-17.40) + 1.73 = 0.01

Joint A1 = M_{C1B1} + $M_{C1C1'}$ = 8.05 – 8.04 = 0.01

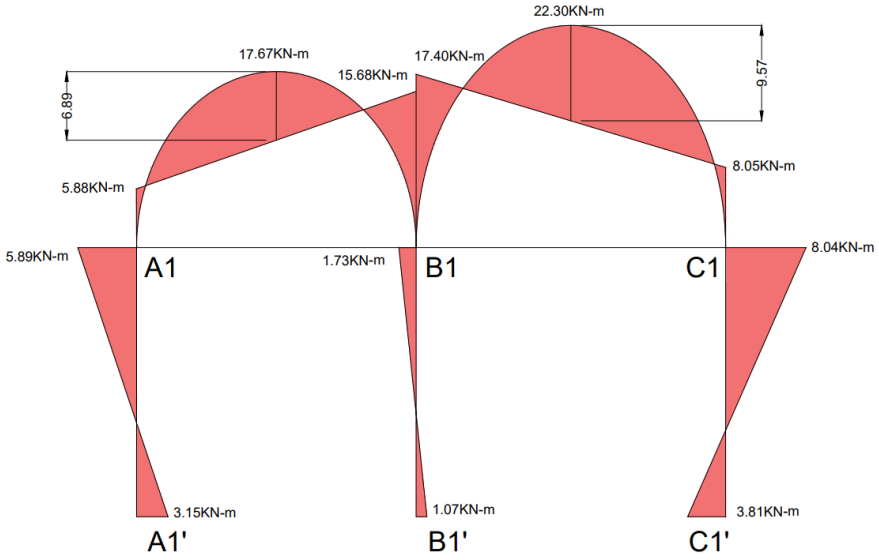


CALCULATION BENDING MOMENT

$BM_{A1B1} = \frac{WL^2}{8} = \frac{18.03 \times 2.8^2}{8} = 17.67KN\text{-}m$

$BM_{B1C1} = \frac{WL^2}{8} = \frac{18.56 \times 3.1^2}{8} = 22.30KN\text{-}m$

BENDING MOMENT DIAGRAM



CALCULATION OF REACTIONS

SPAN A1-B1

Length (L) = 2.8m, UDL (W) = 18.03KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{18.03 \times 2.8}{2} = 25.24KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{B1(Left)} - M_{A1)}}{L} = \frac{15.68 - 5.88}{2.8} = 3.50KN$

Final Reactions:

$R_{A1} = R_{total} - R_{moments} = 25.24 - 3.50 = 21.74KN$

$R_{B1(Left)} = R_{total} + R_{moments} = 25.24 + 3.50 = 28.74KN$

SPAN A1-B1

Length (L) = 3.1m, UDL (W) = 18.56KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{18.56 \times 3.1}{2} = 28.77KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{B1(Right)} - M_{C1)}}{L} = \frac{17.40 - 8.05}{3.1} = 3.02KN$

Final Reactions:

$R_{B1(Right)} = R_{total} + R_{moments} = 28.77 + 3.02 = 31.79KN$

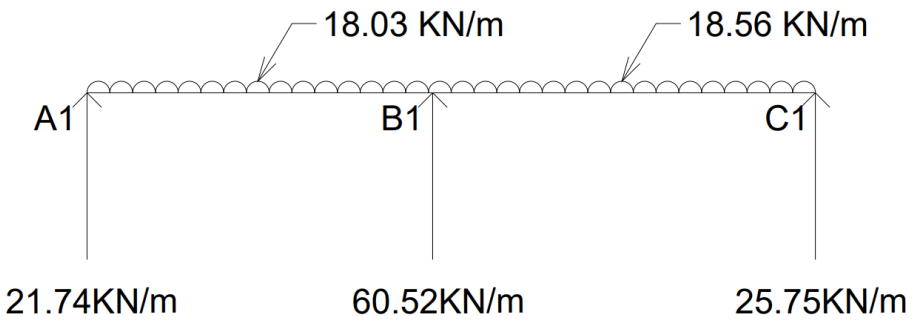
$R_{C1} = R_{total} - R_{moments} = 28.77 - 3.02 = 25.75KN$

Total Reactions:

Reaction at Column A1 (R_{A1}) = 21.74KN

Reaction at Column B1 (R_{B1}) = $R_{B1(Left)} + R_{B1(Right)}$ = 28.74 + 31.78 = 60.52KN

Reaction at Column C1 (R_{C1}) = 25.75KN



CALCULATION OF SHEAR FORCE

SF @ C1 = 25.75KN

SF @ $B1_{(Right)}$ = 25.75 – (18.56 × 3.1) = – 31.78KN

SF @ $B1_{(Right)}$ = 25.75 – (18.56 × 3.1) + 60.52 = 28.74KN

SF @ A1 = 25.75 – (18.56 × 3.1) + 60.52 – (18.03 × 2.8) = –21.74